MODERN

Assessment Question: How has medicine changed from c1250 to the modern day?

Name: ____________________ Class: ____________

Progress & Assessment Tracker	Effort	Level	Teacher Comments
L1: KT4.1: How have modern discoveries changed our understanding of illness?			
L2: KT4.2: How have prevention and treatment advanced in the modern era?			
L3: KT4.3: How was Penicillin discovered and mass-produced?			
L4: KT4.4: How has the government tackled the modern epidemic of Lung Cancer?			
Final Unit Grade:			

Section B: Thematic Study (Medicine in Britain)

> *This section tests your understanding of change, continuity, and causation across 800 years of British medicine.*

Question 3: Explain One Similarity or Difference...

4 Marks | 5 mins

5:00

TARGET OBJECTIVE

AO2 (Analyse similarity and difference across historical periods)

THE SYMMETRICAL SPLICING METHOD

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[Comparative Thesis]   [Era 1: Context]
[Era 2: Matching Context]
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Examiner Red Flags

The "Two-Story" Essay: Writing a block of facts about the first era, followed by a

Grade 9 Checklist

Immediate Comparison: Does my very first sentence make a direct comparison separate block about the second era without linking them is capped at 2 marks.

Concept Slippage: If the question asks about prevention, do not write about treatment.

Chronological Blunders: Placing key developments in the wrong century.

using a comparative connective?

Symmetrical Alignment: Do the details I provide for Era 2 directly match the theme of Era 1?

Single-Paragraph Splicing: Is my answer written as a single, cohesive paragraph?

Question 4: Explain Why... (Causation)

12 Marks | 15-18 mins

18:00

TARGET OBJECTIVE

AO2 (Analyse causation) and AO1 (Demonstrate precise knowledge)

THE THREE-CAUSAL-PILLARS LAYOUT

No Introduction

Para 1	Para 2	Para 3
Stimulus A	Stimulus B	Own Knowledge

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No Conclusion
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Examiner Red Flags

The Stimulus Cap: Failing to introduce an independent third factor from your own knowledge caps your mark at 8/12.

The Narrative Biography Trap: Writing a chronological story instead of explaining *why* their work led to rapid progress.

Grade 9 Checklist

The Rule of Three: Have I structured my answer into exactly three separate paragraphs?

Causal Topic Openers: Does the first sentence of each paragraph state a clear, analytical cause?

Double Causal Connectives: Have I used the Edexcel Connective Chain ("Consequently...", "This meant that...") at least twice per paragraph?

Question 5 & 6: Evaluative Essay

16+4 Marks

25-30 mins

30:00

TARGET OBJECTIVE

AO2 (Evaluate significance/change) and AO1 (Wide-ranging knowledge)

THE GRADE 9 JUDGMENT ARC

1. **Intro** (Thesis & Criteria)
2. **FOR** (Given Factor)
3. **AGAINST** (Own Knowledge)
4. **Conclusion** (Apply Criteria)

Examiner Red Flags

The One-Sided Argument: Failing to analyze alternative factors traps your essay at Level 2 (8 marks).

"Fencing" Conclusions: Reaching a conclusion that simply states both sides were equally important.

Concept Slippage: Treating "care" and "treatment" as identical.

Grade 9 Checklist

Explicit Evaluation Criteria: Have I defined the historical criteria I will use to measure the statement?

The Argument AGAINST: Have I evaluated alternative factors using my own knowledge?

Substantiated Verdict: Have I explicitly applied the criteria established in my intro to justify which factor was most significant?

SPaG: Have I capitalized proper nouns and correctly spelled specialist terms?

L1: KT4.1: How have modern discoveries changed our understanding of illness?

LEARNING OBJECTIVES

- ☐ Explain how the discovery of DNA and the Human Genome Project transformed the understanding of genetic and hereditary illnesses.
- ☐ Analyze the influence of lifestyle choices on modern health and the impact of technology on medical diagnosis.

> *The 20th century witnessed a monumental shift in medicine. With Germ Theory firmly established, scientists soon realised that microbes did not cause every single illness. The groundbreaking discovery of DNA finally unlocked the mystery of hereditary diseases, whilst researchers proved that our own modern lifestyle choices were causing new killer illnesses like cancer. Simultaneously, a technological revolution in diagnosis completely transformed how doctors accurately identified what was wrong with their patients.*

Did you know?

- 1953: The exact year James Watson and Francis Crick successfully discovered the double-helix structure of DNA.
- 1990: The launch year of the Human Genome Project, a massive global scientific effort to decode and map all human DNA.
- 1950: The year the British Medical Research Council published a conclusive study firmly linking smoking cigarettes to lung cancer.

Do Now Activity

Score: / 5

1. What disease did Edward Jenner find a vaccine for in 1796?

2. What did Jenner observe about milkmaids?

3. Who discovered the cause of cholera in 1854?

4. Explain how John Snow proved that cholera was a waterborne disease.

5. To what extent did the government support public health improvements in the mid-19th century?

Vocabulary Check

Fill in the blanks using the vocabulary words below.

DNA | **Human Genome Project** | **Biopsy**

In the twentieth century, diagnosis shifted from a doctor simply observing a patient's symptoms to rigorous laboratory testing. To identify cancers early, doctors today often perform a ________________ on a patient's tissue, examining the sample under powerful microscopes. This transition to evidence-based science was heavily accelerated by breakthroughs in genetics. The 1953 discovery of ________________ proved how traits are passed between generations, which eventually led scientists to launch the massive ________________ to decode and map the complete set of human genes.

KNOWLEDGE CHECK (Q2)

How did Watson and Crick's discovery of the double-helix structure of DNA help scientists search for the causes of hereditary diseases?

☐ A. It provided a complete blueprint of human biology, allowing scientists to identify specific faulty genes responsible for conditions like cystic fibrosis.

☐ B. It proved that hereditary diseases were caused by viruses.

☐ C. It allowed scientists to create antibiotics to kill hereditary diseases.

☐ D. It proved that hereditary diseases were caused by miasma.

Q1. Identify who discovered the double-helix structure of DNA in 1953.

Q3. Describe the goal of the Human Genome Project.

Q4. Explain why Germ Theory could not explain all illnesses.

Q5. Evaluate the impact of the discovery of DNA on modern medicine.

Pair & Share Activity

Prompt: Which modern development has had a more significant impact on our understanding of the causes of illness: the discovery of genetic factors (DNA and hereditary disease) or the scientific identification of lifestyle factors (smoking, diet, and alcohol)?

Think: Jot down your choice and one piece of evidence from the sources (e.g., how mapping the genome allows women to identify risks for breast cancer, versus how smoking is proven to be the biggest cause of preventable diseases in the world, causing heart disease and lung cancer).

Your Notes:

Historian's Corner: The Debate over the Significance of DNA vs. Germ Theory

GCSE historians frequently debate whether the 1953 discovery of the structure of DNA represents a more significant turning point in understanding illness than Pasteur's 1861 Germ Theory. Some historians argue that DNA is the ultimate breakthrough because it finally solved the mystery of non-communicable, hereditary conditions (such as Down's syndrome, haemophilia, and cystic fibrosis) that germs could never explain. However, other historians point out that while Germ Theory immediately transformed surgery, public health, and vaccination in the nineteenth century, the practical impact of DNA has been much slower to develop. Although we can map the genome, genetic treatments like gene therapy are still not widely available for most diseases, meaning the everyday clinical impact of genetics remains a future promise rather than a current reality.

GCSE Exam Practice

Q6. â€˜The discovery of DNA was the most significant breakthrough in understanding the causes of illness in the period c1900â€“present.â€™ How far do you agree? Explain your answer. (16 marks) You may use the following in your answer:
â€¢ Watson and Crick
â€¢ Lifestyle factors
You must also use information of your own.

GCSE Exam Practice

Q Explain why the discovery of the structure of DNA was a significant breakthrough in understanding the causes of illness. (12 marks)

Q Explain one way in which ideas about the cause of illness in the modern period (c1900â€“present) were different from ideas in the period c1700â€“c1900. (4 marks)

Exam Practice

> **1. â€˜The discovery of DNA was the most significant breakthrough in understanding the causes of illness in the period c.1900â€“present.â€™ How far do you agree? Explain your answer. (16 marks)**

You may use the following in your answer:

- Watson and Crick
- Lifestyle factors

You must also use information of your own.

General Notes

☐ Emerging (1-2) ☐ Emerging+ (3) ☐ Expected (4-5) ☐ Expected+ (6-7) ☐ Greater Depth (8-9)

Teacher Comment:

L2: KT4.2: How have prevention and treatment advanced in the modern era?

LEARNING OBJECTIVES

- [ ] Explain how scientific and technological developments led to the creation of magic bullets and antibiotics.
- [ ] Analyze the impact of the National Health Service (NHS) on the provision of and access to medical care.

> *The 20th century transformed medicine from a system that merely managed symptoms into one that actively destroyed diseases. The discovery of targeted chemical cures and powerful antibiotics saved millions of lives from previously fatal infections. Meanwhile, the creation of the NHS revolutionised access to these high-tech treatments, ensuring that for the first time in history, life-saving healthcare and active prevention campaigns were available to everyone, regardless of their wealth.*

Did you know?

- 1909: The year Paul Ehrlich's team discovered Salvarsan 606, the world's very first 'magic bullet' to cure syphilis.
- 1948: The year the National Health Service (NHS) was officially launched, providing free healthcare to all.
- 1956: The year the polio vaccine was introduced in the UK, saving thousands of children from death and paralysis.

Do Now Activity **Score: / 6**

1. What structure did Watson and Crick discover in 1953?

2. What is the function of DNA in relation to disease?

3. Name one modern lifestyle factor that causes disease.

4. Explain how the discovery of DNA changed the understanding of disease.

5. Why did the mapping of the human genome (completed in 2003) not lead to immediate cures?

6. Recall: Who was Roger Bacon and what was their key contribution to medicine?

Vocabulary Check

Write a historically accurate sentence connecting two terms from the glossary box below.

Magic Bullet | **Antibiotics** | **National Health Service (NHS)**

Your Sentence:

KNOWLEDGE CHECK (Q2)

What is the key scientific difference between an antiseptic like carbolic acid and a magic bullet like Salvarsan 606?

☐ A. Antiseptics are used on the outside of the body to prevent infection, whereas magic bullets are injected into the body to target specific microbes.

☐ B. Antiseptics are created from microorganisms, whereas magic bullets are synthetic chemicals.

☐ C. Antiseptics are used to cure viruses, whereas magic bullets cure bacteria.

☐ D. There is no difference; they are both the same thing.

Q1. Identify the first 'Magic Bullet' discovered in 1909.

Q3. Describe how Magic Bullets work.

Q4. Explain how diagnosing illness changed in the 20th century.

Q5. To what extent did the discovery of Magic Bullets revolutionize treatment?

Pair & Share Activity

Prompt: Which development was the most significant turning point in twentieth-century healthcare: the scientific breakthrough of antibiotics, or the political creation of the National Health Service (NHS)?

Think: Jot down your choice and one piece of evidence from the sources (e.g., how penicillin cured previously fatal infections, or how the NHS removed the fear of poverty by providing free healthcare for millions).

Your Notes:

Historian's Corner: The Debate over the Significance of the Beveridge Report in the Foundation of the NHS

Historians debate the extent to which the 1942 Beveridge Report was the primary catalyst for the creation of the NHS in 1948. Traditional consensus histories argue that William Beveridge's report, which recommended tackling the 'five giants' of want, disease, ignorance, squalor, and idleness 'from the cradle to the grave', created an irresistible democratic mandate that forced the post-war Labour government to act. However, revisionist political historians argue that the Beveridge Report was merely a set of recommendations and that the true turning point was the political tenacity of Health Minister Aneurin Bevan. They point out that Bevan had to wage a fierce political battle against the British Medical Association (BMA), who heavily resisted state control, and that the NHS was only realized because Bevan agreed to 'stuff their mouths with gold' by allowing consultants to keep their lucrative private practices alongside their NHS hospital roles.

GCSE Exam Practice

Q6. â€˜Alexander Fleming was the most important individual in the development of penicillin.â€™ How far do you agree? Explain your answer. (16 marks + 4 marks for SPaG)

GCSE Exam Practice

Q Explain why there was rapid progress in the development of penicillin in the 1940s. (12 marks)

Q Explain one way in which the treatment of illness in the modern period (c1900â€“present) was different from the treatment of illness in the period c1700â€“c1900. (4 marks)

Exam Practice

> **1. â€˜Alexander Fleming was the most important individual in the development of penicillin.â€™ How far do you agree? Explain your answer. (16 marks + 4 marks for SPaG) (16 marks)**

You may use the following in your answer:

- Staphylococcus
- Florey and Chain

You must also use information of your own.

General Notes

☐ Emerging (1-2) ☐ Emerging+ (3) ☐ Expected (4-5) ☐ Expected+ (6-7) ☐ Greater Depth (8-9)

Teacher Comment:

L3: KT4.3: How was Penicillin discovered and mass-produced?

LEARNING OBJECTIVES

- [ ] Contrast the roles of Alexander Fleming with those of Howard Florey and Ernst Chain in the penicillin breakthrough.
- [ ] Analyze how the factors of government intervention, science and technology, and war collaborated to enable the mass production of penicillin.

> *Before the 20th century, a simple scratch or an infected wound could easily lead to fatal blood poisoning. The discovery of penicillin, the world's very first antibiotic, marked a monumental turning point in modern medicine. Driven by the desperate, bloody necessity of the Second World War, scientists finally managed to harness naturally occurring microbes to actively destroy bacterial infections, saving millions of lives and permanently transforming modern healthcare. * The_Penicillin_Revolution.pptx19mb*

Did you know?

- 1928: The exact year Alexander Fleming accidentally discovered penicillin mould killing bacteria in his unwashed petri dish.
- 1941: The year Florey and Chain successfully treated their first human patient, Albert Alexander, proving the purified drug worked.
- 15%: The estimated percentage of injured Allied soldiers whose lives were saved by mass-produced penicillin during WWII.

Do Now Activity **Score: / 6**

1. What is a 'magic bullet'?

2. Name the first magic bullet discovered by Paul Ehrlich in 1909.

3. What was established in Britain in 1948 to improve public health?

4. Explain the impact of the NHS on medical treatment in Britain.

5. How did technology improve diagnosis in the 20th century?

6. Recall: Who was Ernst Chain and what was their key contribution to medicine?

Vocabulary Check

Write a clear definition and a historically accurate sentence for the term: ***Antibiotic***

Q1. Identify who discovered the first antibiotic in 1928.

Q2. Describe how Penicillin was discovered.

KNOWLEDGE CHECK (Q5)

What was the main scientific and practical limitation that caused Fleming to abandon his work on penicillin in 1931?

- ☐ A. He realized that penicillin was highly toxic to humans.
- ☐ B. He could not secure funding from the government because it was peacetime.
- ☐ C. He lacked the chemical expertise and facilities to extract and purify penicillin in large enough quantities for human trials.
- ☐ D. He was drafted into the army to fight in World War II.

Q3. Explain the roles of Florey and Chain in the development of Penicillin.

Q4. Evaluate the role of World War II in the mass production of Penicillin.

Pair & Share Activity

Prompt: Who deserves the most credit for the penicillin breakthrough: Alexander Fleming for his accidental discovery in 1928, or Howard Florey and Ernst Chain for their scientific development and therapeutic isolation of the drug in the 1940s?

Think: Jot down your choice and one piece of evidence from the sources (e.g., Fleming recognizing and publishing the mould's bacteria-killing properties in 1928, or the Oxford team actually turning it into a usable drug and proving its clinical effectiveness on humans).

Your Notes:

Historian's Corner: The Debate over the 'Myth' of Fleming and the Penicillin Breakthrough

For decades, popular historical accounts portrayed Alexander Fleming as the sole hero of the penicillin story, celebrating his accidental observation of the Penicillium mould in 1928 as an instant cure. However, revisionist historians, such as Patricia Harding, actively challenge this traditional narrative, arguing that it propagates a 'myth'. They point out that Fleming actually abandoned his work on penicillin by 1931, having failed to isolate the active substance or prove that it could kill bacteria in living human blood. Revisionists argue that the true breakthrough belongs to Howard Florey, Ernst Chain, and the Oxford research team, who systematically isolated the drug in 1940 and proved its therapeutic value, but were initially ignored by a public captivated by the romantic narrative of Fleming's accident.

GCSE Exam Practice

Q6. Explain why there was rapid progress in disease prevention in the period c1900â€“present. (12 marks)

GCSE Exam Practice

Q Explain why the government took a more active role in preventing disease in the period c1900â€“present. (12 marks)

Q Explain one way in which ideas about the prevention of illness in the modern period (c1900â€“present) were different from the medieval period (c1250â€“c1500). (4 marks)

Exam Practice

> **1. Explain why there was rapid progress in disease prevention in the period c1900â€“present. (12 marks)**

You may use the following in your answer:

- Government vaccination campaigns
- Legislation

You must also use information of your own.

General Notes

☐ Emerging (1-2) ☐ Emerging+ (3) ☐ Expected (4-5) ☐ Expected+ (6-7) ☐ Greater Depth (8-9)

Teacher Comment:

L4: KT4.4: How has the government tackled the modern epidemic of Lung Cancer?

LEARNING OBJECTIVES

- ☐ Explain how modern science and technology have transformed the diagnosis and treatment of lung cancer since c1900.
- ☐ Analyze the effectiveness and changing focus of government legislation to prevent smoking-related lung cancer.

> *Lung cancer was incredibly rare in the nineteenth century but rapidly became the second most common cancer in the UK by the twenty-first century. After the link between smoking and cancer was conclusively proven in 1950, medicine had to develop advanced high-tech methods to diagnose and treat this deadly disease. Simultaneously, the crisis forced the government to abandon its reluctance to interfere in personal habits, leading to aggressive modern prevention campaigns.*

Did you know?

- 1950: The year the British Medical Research Council published its conclusive study linking cigarette smoking to lung cancer.
- 2007: The year the government passed a law banning smoking in all enclosed public workplaces and raised the legal age for buying tobacco to 18.
- 85%: The approximate percentage of lung cancer cases that are caused by people who smoke or have previously smoked.

Do Now Activity

Score: / 6

1. Who discovered Penicillin by accident in 1928?

2. Which two scientists developed Penicillin into a usable drug?

3. What role did the US government play in the mass production of Penicillin?

4. Explain why Penicillin is considered a turning point in medicine.

5. To what extent was Fleming solely responsible for the success of Penicillin?

6. Recall: Who was Roger Bacon and what was their key contribution to medicine?

Vocabulary Check

Fill in the blanks using the vocabulary words below.

Bronchoscopy | **Immunotherapy** | **Nanny State**

In twenty-first-century Britain, the diagnosis and treatment of lung cancer have been transformed by science and technology. Rather than relying on simple chest X-rays, doctors can perform a highly precise ________________ to examine the inside of the lungs and collect cell samples for laboratory testing. When tumours are detected, patients are often treated with cutting-edge therapies such as ________________ to trigger their bodyâ€™s natural defenses to destroy cancer cells. At the same time, the government has passed highly restrictive laws to prevent people from smoking, leading some critics to complain that state interference in lifestyle choices represents the rise of an overbearing ________________.

Q1. Identify one method used to diagnose Lung Cancer.

Q2. Describe the aggressive treatments used for Lung Cancer.

Q3. Explain why the UK government has focused heavily on preventing Lung Cancer.

KNOWLEDGE CHECK (Q5)

What is the key clinical advantage of using a CT scan over a traditional chest X-ray when diagnosing a lung cancer patient?

☐ A. A CT scan uses harmless sound waves instead of radiation.

☐ B. A CT scan can be performed at home by the patient.

☐ C. A CT scan creates highly detailed, 3D cross-sectional images of the lungs, allowing doctors to detect much smaller tumors than a 2D X-ray.

☐ D. A CT scan can actively cure the lung cancer by destroying the tumor.

Q4. Evaluate the effectiveness of government legislation in reducing smoking.

Pair & Share Activity

Prompt: Which category of modern government action has been the most significant turning point in the prevention of lung cancer: directly warning and educating current smokers (such as plain packaging and graphic warning labels), preventing young people from starting (such as raising the legal age of sale from 16 to 18), or protecting non-smokers from second-hand smoke (such as the 2007 indoor smoking ban)?

Think: Jot down your choice and one piece of evidence from your knowledge (for instance, the 2007 ban significantly shifting social attitudes about where smoking is acceptable, or how 1950s research first proved the definitive link between tobacco and cancer).

Your Notes:

Historian's Corner: The Debate over the Motives for Modern Public Health Interventions

Historians of modern medicine actively debate what primarily drove the British government to finally abandon its traditional hands-off approach and launch aggressive campaigns against smoking. Traditionalist historians argue that the state acted out of genuine humanitarian concern and scientific responsibility, moving to intervene once Richard Doll and A. Bradford Hill's landmark 1950 study provided undeniable statistical proof linking tobacco to lung cancer. Conversely, revisionist historians argue that the government's response was actually incredibly slow and reluctant, delayed for decades by a desire to protect massive tobacco tax revenues and a fear of political backlash against state overreach. These historians suggest that the real turning point was economic pragmatism: as the financial burden of treating chronic, smoking-related illnesses on the newly established National Health Service (NHS) became *unsustainable, the government was forced to recognize that prevention was far cheaper than clinical treatment.*

GCSE Exam Practice

Q6. Explain why the government took action to improve public health in the period c1900â€“present. (12 marks)

GCSE Exam Practice

Q Explain why the creation of the National Health Service (NHS) in 1948 improved public health in Britain. (12 marks)

Q Explain one way in which hospital care in the modern period (c1900â€“present) was different from hospital care in the period c1700â€“c1900. (4 marks)

Exam Practice

> **1. Explain why the government took action to improve public health in the period c1900â€“present. (12 marks)**

You may use the following in your answer:

- Lung cancer
- The NHS

You must also use information of your own.

General Notes

☐ Emerging (1-2) ☐ Emerging+ (3) ☐ Expected (4-5) ☐ Expected+ (6-7) ☐ Greater Depth (8-9)

Teacher Comment:

Factors Overview: Modern

Edexcel focuses heavily on the factors that drove medical progress (or held it back). For each factor below, write one specific historical example from this period that either helped or hindered medical progress.

Factor	Specific Historical Example & Impact
Individuals	
The Church & Religion	
Government & Wealth	
Science & Technology	
Attitudes in Society	
War	

Factor	Specific Historical Example & Impact